Assignment-2

Static Website:

Static websites are the websites that doesn't change the content or layout dynamically with every request to the web server. Static websites display exactly the same information whenever anyone visits it. User sees the updated content of Static Website only when a web author manually updates them with a **text editor** or any web editing tool used for creating websites.

- Static websites are the simplest kind of website that you can build.
- Every viewer will see the exactly same text, multimedia design or video every time he/she visits the website until you alter that page's source code.
- Static websites are written with the help of **HTML and CSS.**
- The only form of interactively on a static website is **hyperlink**.
- Static website can be used for the information that doesn't change substantially over months or even years.
- Static pages are easy and simple to understand, secure, less prone to technology errors and breakdown and easily visible by search engines.
- **HTML** was the first tool with which people had begun to create static web pages.
- Static websites provide **flexibility**.
- **Lightweight** .
- Static websites perform **faster** and **well** than dynamic ones.

Dynamic Website

Dynamic Website is a website containing data that can be mutable or changeable. It uses client-side or server scripting to generate mutable content.

A dynamic website is more interactive than a static website and can change based on user input or real-time data. Imagine an e-commerce store where product prices and availability fluctuate—this requires a dynamic website to display the most up-to-date information.

IP Address (Internet Protocol)

An IP Address (Internet Protocol Address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication. It serves two main purposes:

1. Identifying a device on the network.
2. Locating the device to enable communication with other devices over a network like the Internet.

DNS (Domain Name System)

The Domain Name System (DNS) turns domain names into IP addresses, which browsers use to load internet pages. Every device connected to the internet has its own IP address, which is used by other devices to locate the device. DNS servers make it possible for people to input normal words into their browsers, such as Fortinet.com, without having to keep track of the IP address for every website.

HTTP and HTTPS (Hyper Text Transfer Protocol and Hyper Text Transfer Protocol Secure)

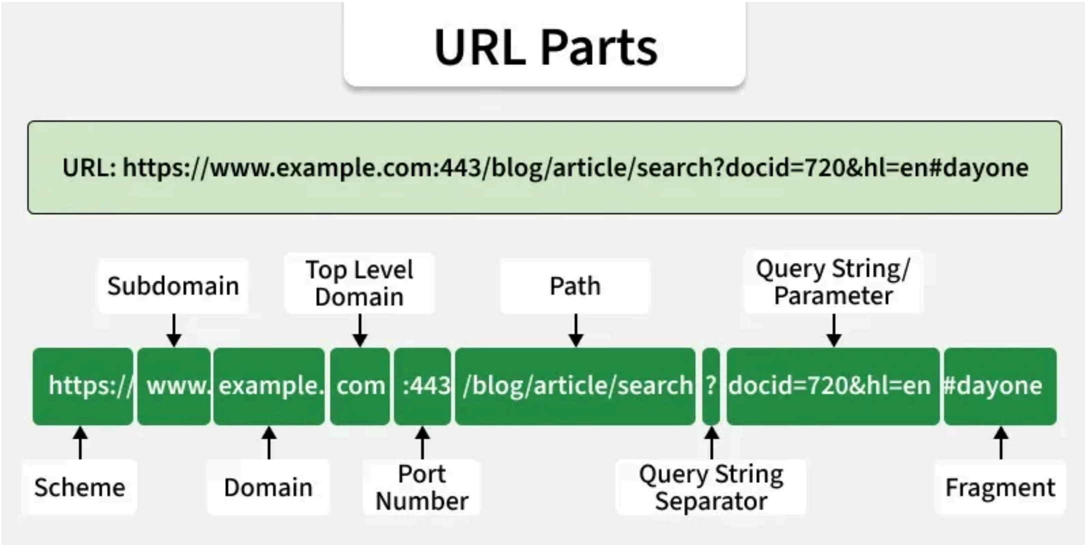
the foundational systems that allow your web browser to communicate with websites.

HTTP: Sends all data in plain text. If you type a password or credit card number into an HTTP website, anyone intercepting your network traffic can read that information. Modern browsers like Google Chrome will often flag HTTP sites as "Not Secure" in the address bar.

HTTPS: Is the secure, encrypted version of HTTP. It uses SSL (Secure Sockets Layer) or TLS (Transport Layer Security) protocols to scramble your data before it is transmitted. This ensures that even if an attacker intercepts your connection, they cannot decipher what you are doing.

Uniform Resource Locator (URL):

A URL (Uniform Resource Locator) is a unique web address used to identify and access resources available on the internet. It helps users locate web pages, files, and online services through a browser.


URL Parts
URL: https://www.example.com:443/blog/article/search?docid=720&hl=en#dayone
Subdomain
Top Level Domain
Path
Query String/ Parameter
https://
www.
example.
com
:443
/blog/article/search
?
docid=720&hl=en
#dayone
Scheme
Domain
Port Number
Query String Separator
Fragment